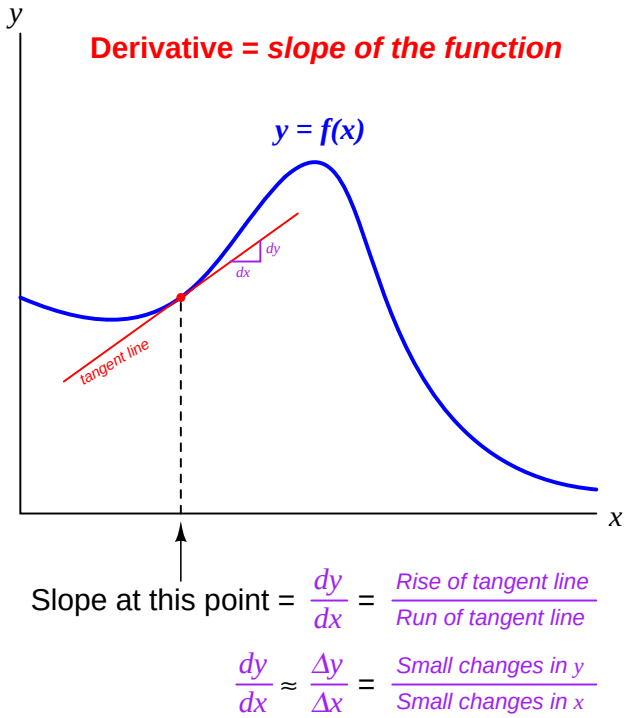
1.4 How derivatives and integrals relate to one another

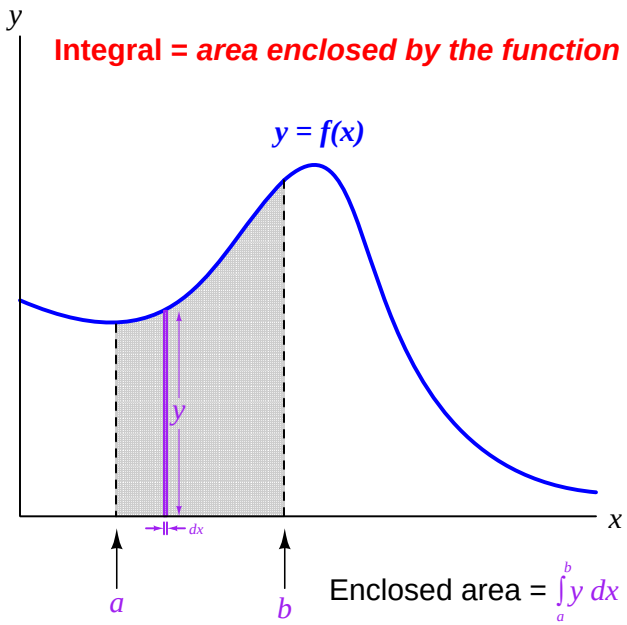
First, let us review some of the properties of *differentials* and *derivatives*, referencing the expression and graph shown below:

- A *differential* is an infinitesimal increment of change (difference) in some continuously-changing variable, represented either by a lower-case Roman letter d or a lower-case Greek letter "delta" (δ). Such a change in time would be represented as dt; a similar change in temperature as dT; a similar change in the variable x as dx.
- A derivative is always a *quotient of differences*: a process of subtraction (to calculate the amount each variable changed) followed by division (to calculate the *rate* of one change to another change).
- The units of measurement for a derivative reflect this final process of division: one unit divided by some other unit (e.g. gallons *per* minute, feet *per* second).
- Geometrically, the derivative of a function is its graphical *slope* (its "rise over run").
- When computing the value of a derivative, we must specify a single point along the function where the slope is to be calculated.
- The *tangent line* matching the slope at that point has a "rise over run" value equal to the derivative of the function at that point.



Next, let us review some of the properties of *integrals*, referencing the expression and graph shown below:

- An integral is always a *sum of products*: a process of multiplication (to calculate the product of two variables) followed by addition (to sum those quantities into a whole).
- The units of measurement for an integral reflect this initial process of multiplication: one unit times some other unit (e.g. kilowatt-hours, foot-pounds, volt-seconds).
- When computing the value of an integral, we must specify both the starting and ending points along the function defining the interval of integration (a and b).
- Geometrically, the integral of a function is the graphical *area* enclosed by the function and the interval boundaries.
- The area enclosed by the function may be thought of as an infinite sum of extremely narrow rectangles, each rectangle having a height equal to one variable (y) and a width equal to the differential of another variable (dx).



Upper boundary of area = y
Lower boundary of area = 0
Starting boundary of area = a
Ending boundary of area = b

Just as division and multiplication are *inverse* mathematical functions (i.e. one "un-does" the other), differentiation and integration are also inverse mathematical functions. The two examples of propane gas flow and mass measurement highlighted in the previous sections illustrates this complementary relationship. We may use differentiation with respect to time to convert a mass measurement (m) into a mass flow measurement (W, or $\frac{dm}{dt}$). Conversely, we may use integration with respect to time to convert a mass flow measurement (W, or $\frac{dm}{dt}$) into a measurement of mass gained or lost (Δm).

Likewise, the common examples of position (x), velocity (v), and acceleration (a) used to illustrate the principle of differentiation are also related to one another by the process of integration. Reviewing the derivative relationships:

$$v = \frac{dx}{dt} \qquad \text{Velocity is the derivative of position with respect to time}$$

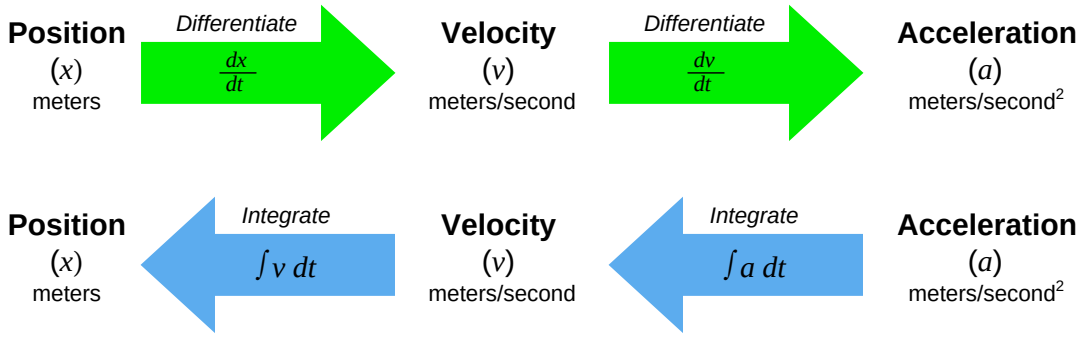
$$a = \frac{dv}{dt} \qquad \text{Acceleration is the derivative of velocity with respect to time}$$

Now, expressing position and velocity as *integrals* of velocity and acceleration, respectively[9]:

$$x = \int v\, dt \qquad \text{Position is the integral of velocity with respect to time}$$

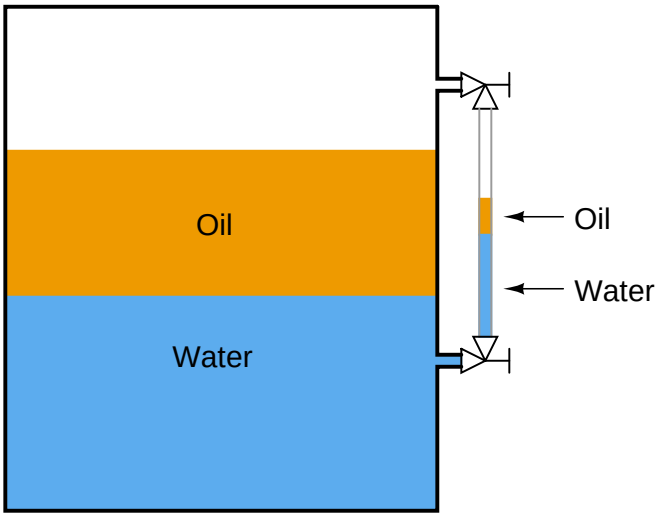
$$v = \int a\, dt \qquad \text{Velocity is the integral of acceleration with respect to time}$$

Differentiation and integration may be thought of as processes transforming these quantities into one another. Note the transformation of units with each operation – differentiation always divides while integration always multiplies:

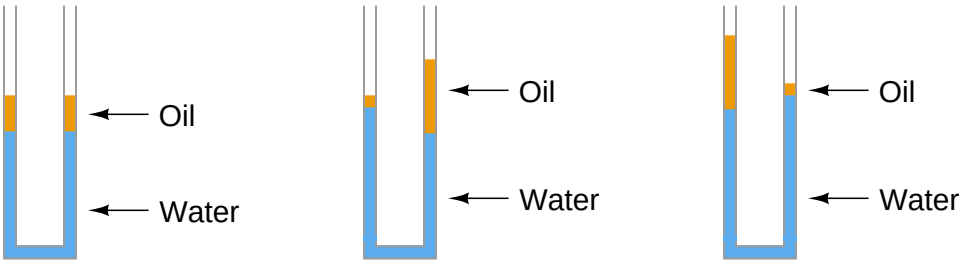


[9] To be perfectly accurate, we must also include initial values for position and velocity. In other words, $x = \int v dt + x_0$ and $v = \int a\, dt + v_0$

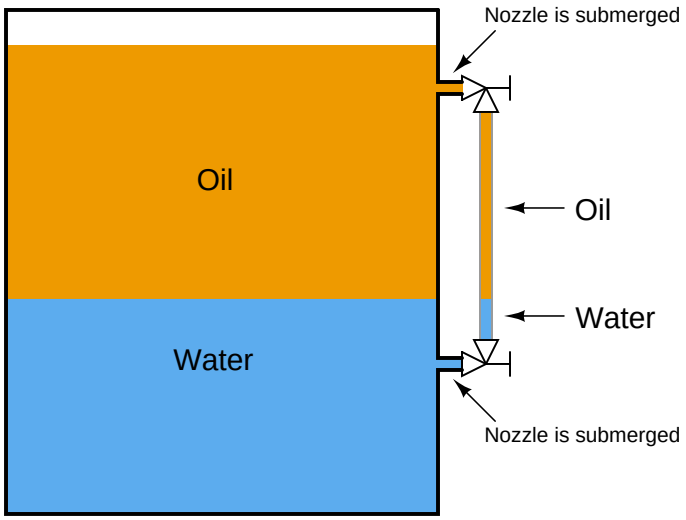
If by chance some oil does find its way into the sightglass tube – either by the interface level dropping below the lower nozzle or the total level rising above the upper nozzle – the oil/water interface shown inside the level gauge may not continue to reflect the true interface inside the vessel once the interface and total levels return to their previous positions:



Recall that the level gauge and vessel together form a U-tube manometer. So long as the pressures from each liquid column are the same, the columns balance each other. The problem is, many different liquid-liquid interface columns can have the same hydrostatic pressure without being identical to one another:

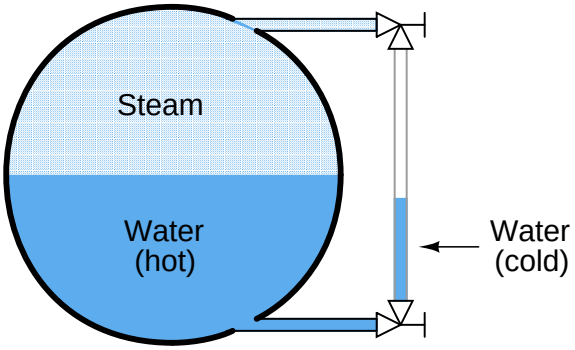


The only way to ensure proper two-part liquid interface level indication in a sightglass is to keep both ports (nozzles) submerged:



20.1.3 Temperature problems

Another troublesome scenario for level gauges is when the liquid inside the vessel is substantially hotter than the liquid in the gauge, causing the densities to be different. This is commonly seen on boiler level gauges, where the water inside the sightglass cools off substantially from its former temperature inside the boiler drum:

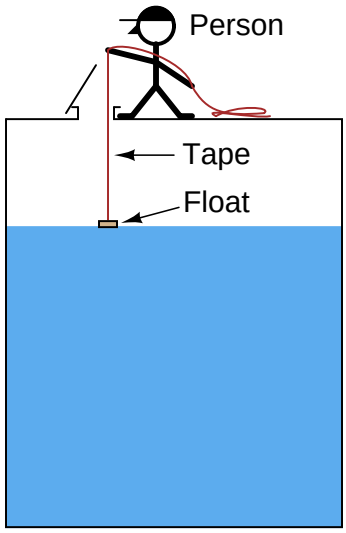


Looking at the sightglass as a U-tube manometer again, we see that unequal-height liquid columns may indeed balance each other's hydrostatic pressures if the two columns are comprised of liquids with different densities. The weight density of water is 62.4 lb/ft^3 at standard temperature, but may be as low as only 36 lb/ft^3 at temperatures common for power generation boilers.

20.2 Float

Perhaps the simplest form of solid or liquid level measurement is with a *float*: a device that rides on the surface of the fluid or solid within the storage vessel. The float itself must be of substantially lesser density than the substance of interest, and it must not corrode or otherwise react with the substance.

Floats may be used for manual "gauging" of level, as illustrated here:



A person lowers a float down into a storage vessel using a flexible measuring tape, until the tape goes slack due to the float coming to rest on the material surface. At that point, the person notes the length indicated on the tape (reading off the lip of the vessel access hole). This distance is called the *ullage*, being the distance from the top of the vessel to the surface of the process material. *Fillage* of the vessel may be determined by subtracting this "ullage" measurement from the known height of the vessel.

Obviously, this method of level measurement is tedious and may pose risk to the person conducting the measurement. If the vessel is pressurized, this method is simply not applicable.

flowmeters work on the principle of convective cooling, this means a fluid having a high specific heat value will elicit a greater response from a thermal mass flowmeter than the exact same mass flow rate of a fluid having a lesser specific heat value (i.e. a fluid that is not as good of a coolant).

This means we must know the specific heat value of whatever fluid we plan to measure with a thermal mass flowmeter, and we must be assured its specific heat value will remain constant. For this reason, thermal mass flowmeters are not suitable for measuring the flow rates of fluid streams whose chemical composition is likely to change over time. This limitation is analogous to that of a pressure sensor used to hydrostatically measure the level of liquid in a vessel: in order for this level-measurement technique to be accurate, we must know the density of the liquid and also be assured that density will be constant over time.

Thermal mass flowmeters are simple and reliable instruments. While not as accurate or tolerant of piping disturbances as Coriolis mass flowmeters, they are far less expensive.

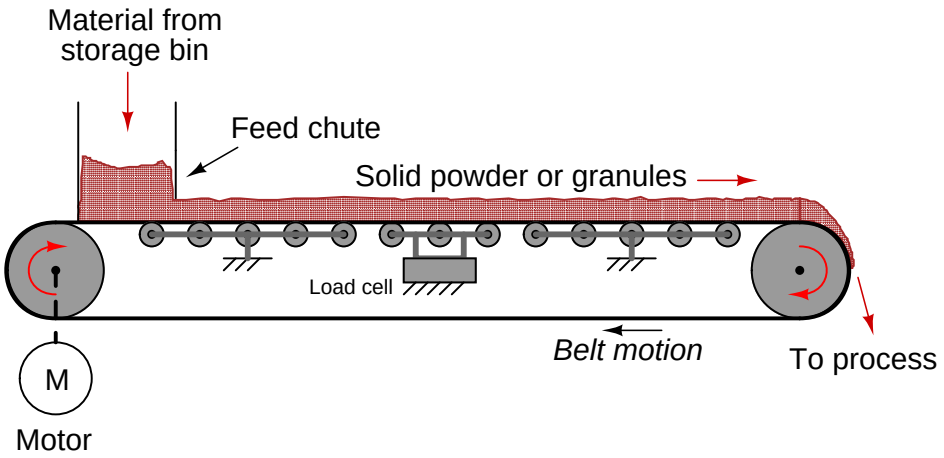
Perhaps the greatest disadvantage of thermal mass flowmeters is their sensitivity to changes in the specific heat of the process fluid. This makes the calibration of any thermal mass flowmeter specific for one composition of fluid only. In some applications such as automotive engine intake air flow, where the fluid composition is constant, this limitation is not a factor. In many industrial applications, however, this limitation is severe enough to prohibit the use of thermal mass flowmeters. Industrial applications for thermal mass flowmeters include natural gas flow measurement (non-custody transfer), and the measurement of purified gas flows (oxygen, hydrogen, nitrogen) where the composition is known to be very stable.

Another (potential) limitation of thermal mass flowmeters is the sensitivity of some designs to changes in flow regime. Since the measurement principle is based on heat transfer by fluid convection, any factor influencing the convective heat-transfer efficiency will translate into a perceived difference in mass flow rate. It is a well-known fact in fluid mechanics that turbulent flows are more efficient at convecting heat than laminar flows, because the "stratified" nature of a laminar flowstream impedes heat transfer across the fluid width[76]. In some thermal flowmeter designs, the walls of a heated metal *tube* serve as the "hot" element cooled by the fluid, and the difference between the rate of heat transferred by a laminar flowstream from the walls of a heated tube versus a turbulent flowstream can be great. Therefore, a change in flow regime (from turbulent to laminar, and vice-versa) will cause a calibration shift for this design of thermal mass flowmeter.

[76] In a laminar flowstream, individual molecules do not cross paths, but rather travel in parallel lines. This means only those molecules traveling near the wall of a tube will be exposed to the temperature of the wall. The lack of "mixing" in a laminar flowstream means molecules traveling in the inner portions of the stream never contact the tube wall, and therefore never serve to transfer heat directly to or from the wall. At best, those inner-path molecules transfer heat by *conduction* with adjacent molecules which is a less efficient transfer mechanism than convection.

22.8 Weighfeeders

A special type flowmeter suited for powdered or granular solids is the *weighfeeder*. One of the most common weighfeeder designs consists of a conveyor belt with a section supported by rollers coupled to one or more load cells, such that a fixed length of the belt is continuously weighed:



The load cell measures the weight of a fixed-length belt section, yielding a figure of material weight per linear distance on the belt. A tachometer (speed sensor) measures the speed of the belt. The product of these two variables is the mass flow rate of solid material "through" the weighfeeder:

$$W = \frac{Fv}{d}$$

Where,

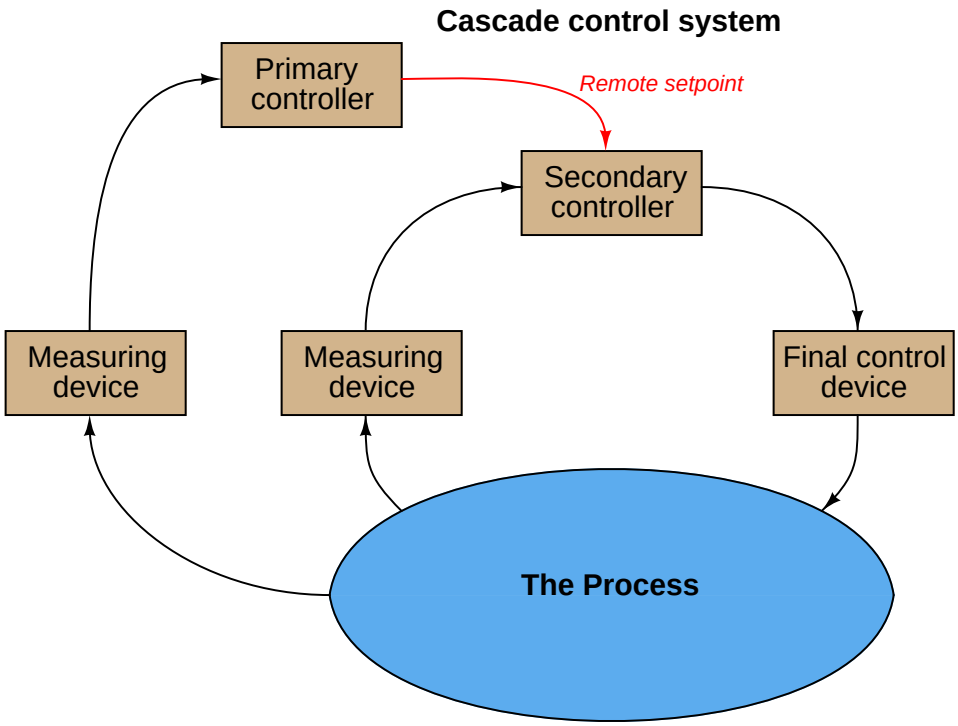
W = Mass flow rate (e.g. pounds per second)

F = Force of gravity acting on the weighed belt section (e.g. pounds)

v = Belt speed (e.g. feet per second)

d = Length of weighed belt section (e.g. feet)

Thus, a cascade control system consists of two feedback control loops, one nested inside the other:



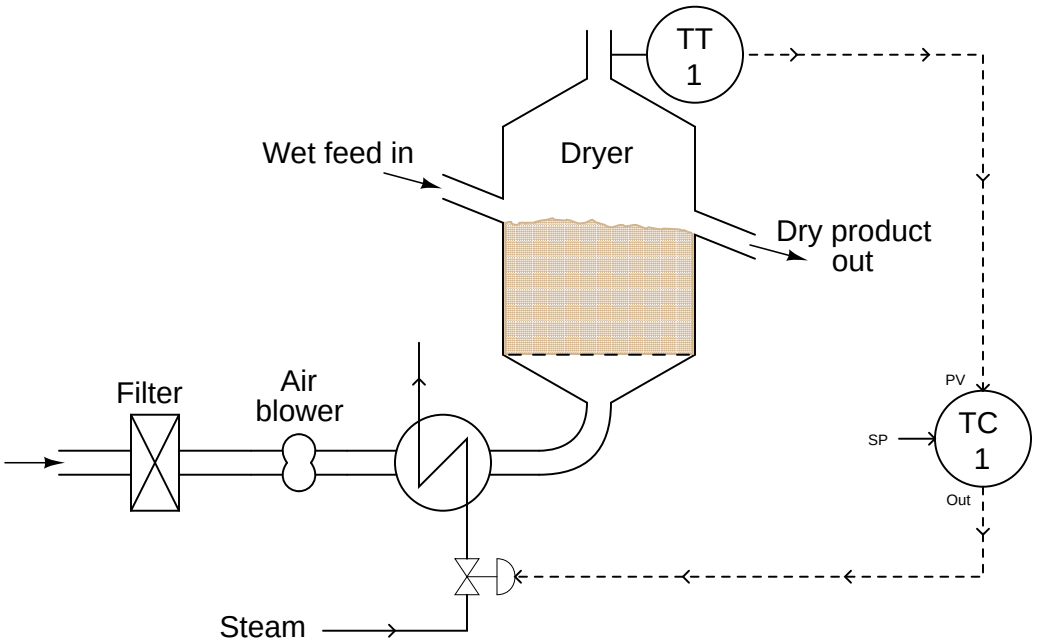
A very common example of cascade control is a *valve positioner*, which receives a command signal from a regular process controller, and in turn works to ensure the valve stem position precisely matches that command signal. The control valve's stem position is the process variable (PV) for the positioner, just as the command signal is the positioner's setpoint (SP). Valve positioners therefore act as "slave" controllers to "master" process controllers controlling pressure, temperature, flow, or some other process variable.

The purpose of cascade control is to achieve greater stability of the primary process variable by regulating a secondary process variable in accordance with the needs of the first. An essential requirement of cascaded control is that the secondary process variable be faster-responding (i.e. shorter lag and dead times) than the primary process variable.

An analogy for understanding cascade control is that of *delegation* in a work environment. If a supervisor delegates some task to a subordinate, and that subordinate performs the task without further need of guidance or assistance from the supervisor, the supervisor's job is made easier. The subordinate takes care of all the little details that would otherwise burden the supervisor if the supervisor had no one to delegate to. This analogy also makes it clear why the secondary process variable must be faster-responding than the primary process variable: the supervisor-subordinate management structure fails to work if the supervisor does not maintain focus on long-term goals (i.e. longer-term than the completion time of the tasks given to subordinates). If a supervisor focuses on achieving goals that are shorter-term than the time required for subordinates to complete

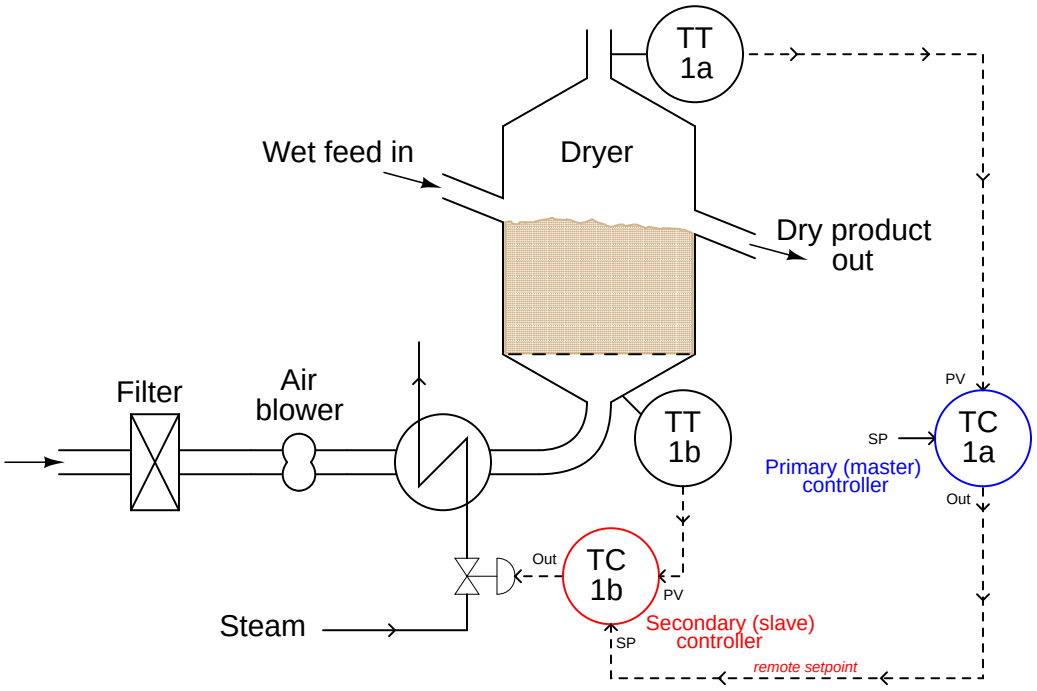
their assignments, the supervisor will inevitably call for "course changes" that are too quick for the subordinates to execute. This will lead to the subordinates "lagging" behind the supervisor's orders, to the detriment of everyone's satisfaction.

An example of cascade control applied to a real industrial process is shown here, for a *dryer* system where heated air is used to evaporate water from a granular solid. The primary process variable is the outlet air exiting the dryer, which should be maintained at a high enough temperature to ensure water will not remain in the upper layers of the solid material. This outlet temperature is fairly slow to react, as the solid material mass creates a large lag time:



There are several parameters influencing the temperature of the outlet air other than the moisture content of the drying material. These include air flow, ambient air temperature, and variations in steam temperature. Each one of these variables is a *load* on the process variable we are trying to control (outlet air temperature). If any of these parameters were to suddenly change, the effect would be slow to register at the outlet temperature even though there would be immediate impact at the bottom of the dryer where the heated air enters. Correspondingly, the control system would be slow to correct for any of these changing loads.

We may better compensate for these loads by installing a second temperature transmitter at the inlet duct of the dryer, with its own controller to adjust steam flow at the command of the primary controller:



Now, if any of the loads related to incoming air flow or temperature vary, the secondary controller (TC-1b) will *immediately* sense the change in dryer inlet temperature and compensate by adjusting steam flow through the heat exchanger. Thus, the "slave" control loop (1b) helps stabilize the "master" control loop (1a) by reacting to load changes long before any effect might manifest at the dryer outlet.

A helpful way to think of this is to consider the slave controller as *shielding* the master controller from the loads previously mentioned (incoming air flow, ambient temperature, and steam temperature). Of course, these variables still act as loads to the slave controller, as it must continuously adjust the steam valve to compensate for changes in air flow, ambient air temperature, and steam temperature. However, so long as the slave controller does a good job of stabilizing the air temperature entering the dryer, the master controller will never "see" the effects of those load changes. Responsibility for incoming air temperature has been delegated to the slave controller, and as a result the master controller is conveniently isolated from the loads impacting that loop.

To re-emphasize an important point, one of the non-negotiable requirements for cascade control is that the secondary (slave) loop must be *faster-responding* than the primary (master) loop. Cascade control cannot function if this speed relationship is reversed. Temperature controller TC-1b is able to be a slave to controller TC-1a because the natural response time of the temperature at the dryer's bottom is much shorter than at the dryer's top with respect to any changes in steam valve position.